Homework

DM13 - Bookings and overbooking - 60 min.

A room has eighteen seats. The organiser accepts twenty bookings, paid in advance.
Each person who has booked turns up with probability 0 ,8, independently of the others.
Let X be the number of people who turn up. If more than eighteen people come, the extra people are turned away.

1. Justify the distribution of X, then calculate its expectation and variance.

2. Calculate the probability that the organiser must turn away at least one person. Round to 10^{-6}.

3. Each booking earns 10 monetary units, even if the person does not come. Each person turned away receives a total payment of 30 monetary units, refund and compensation included. Let $R = \max(X - 18; 0)$ be the number of people turned away. Express the gain G in terms of R, then calculate $E(G)$.

4. Calculate the gain if the organiser accepts only eighteen bookings, with the same payment rules. Compare the two strategies in terms of mean gain.

5. Name one model assumption that may be debatable in practice, then a criterion other than the monetary gain to be taken into account in this decision.